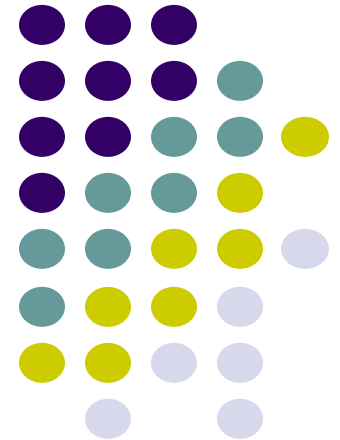


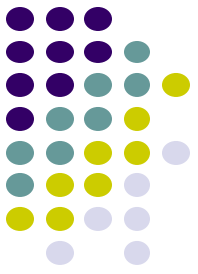
ErSE222: Machine learning in Geoscience

March. 12th, 2025

Tariq Alkhalifah and Omar Saad



Skip connections



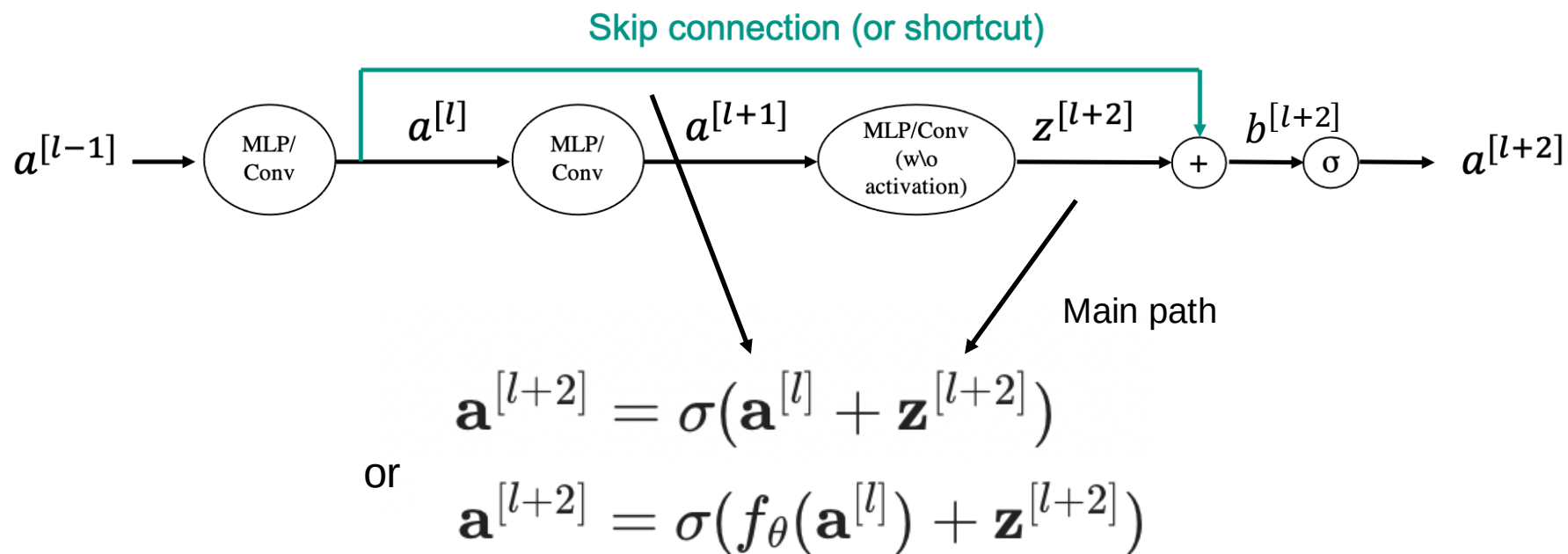
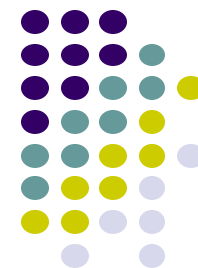
Deep neural networks can learn more complex relationships...

... but they are prone to slow training due to **vanishing gradients**

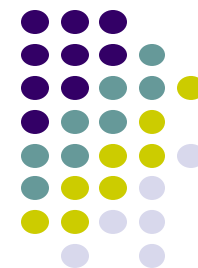
Skip connections: introduced in 2015 under the name of **Residual blocks**

take the activation of a certain layer and feed it directly to another layer further down in the computational graph.

Skip connections



Skip connections



Why successful?

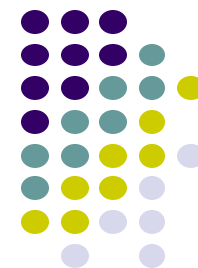
$$\mathbf{b}^{[l+2]} = \mathbf{a}^{[l]} + \mathbf{z}^{[l+2]} = \mathbf{a}^{[l]} + f_{l+2}(f_{l+1}(\mathbf{a}^{[l]}))$$

$$\mathbf{a}^{[l+2]} = \sigma(\mathbf{b}^{[l+2]})$$

$$\frac{\partial \mathbf{b}^{[l+2]}}{\partial \mathbf{a}^{[l]}} = 1 + \frac{\partial \mathbf{a}^{[l+1]}}{\partial \mathbf{a}^{[l]}} \frac{\partial \mathbf{z}^{[l+2]}}{\partial \mathbf{a}^{[l+1]}}$$

ensures that gradient is never 0!

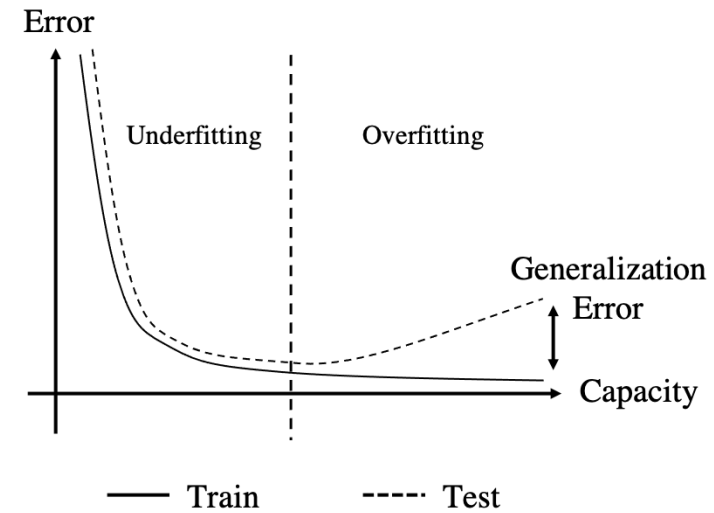
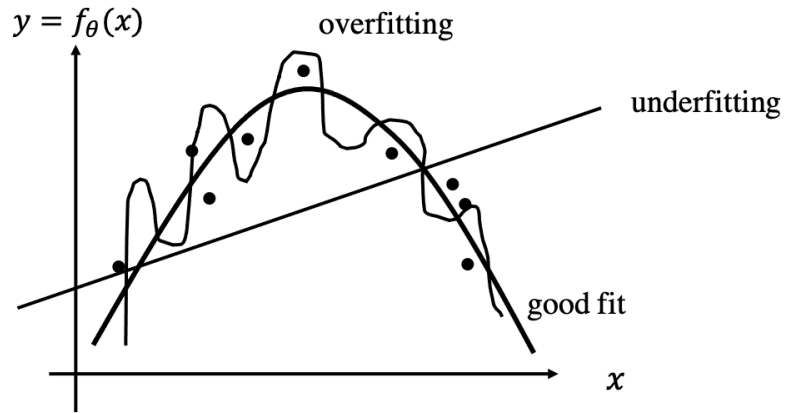
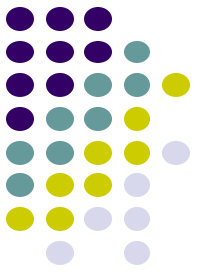
ML performance



Two different (usually contrasting) goals:

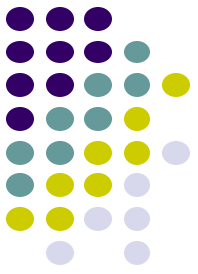
- **Bias:** ability to produce a small training error. When the error is small, we say that we have a model with low bias. Vice versa, when the error is large, the model have high bias. In this case, the model is said to be *underfitting* the data;
- **Variance:** ability to produce comparable training and testing error. In other words, if we define *gap* to be the difference between the testing and training errors, this is also the ability to produce a small gap. When the gap is large, the model is said to be *overfitting* the data.

ML performance

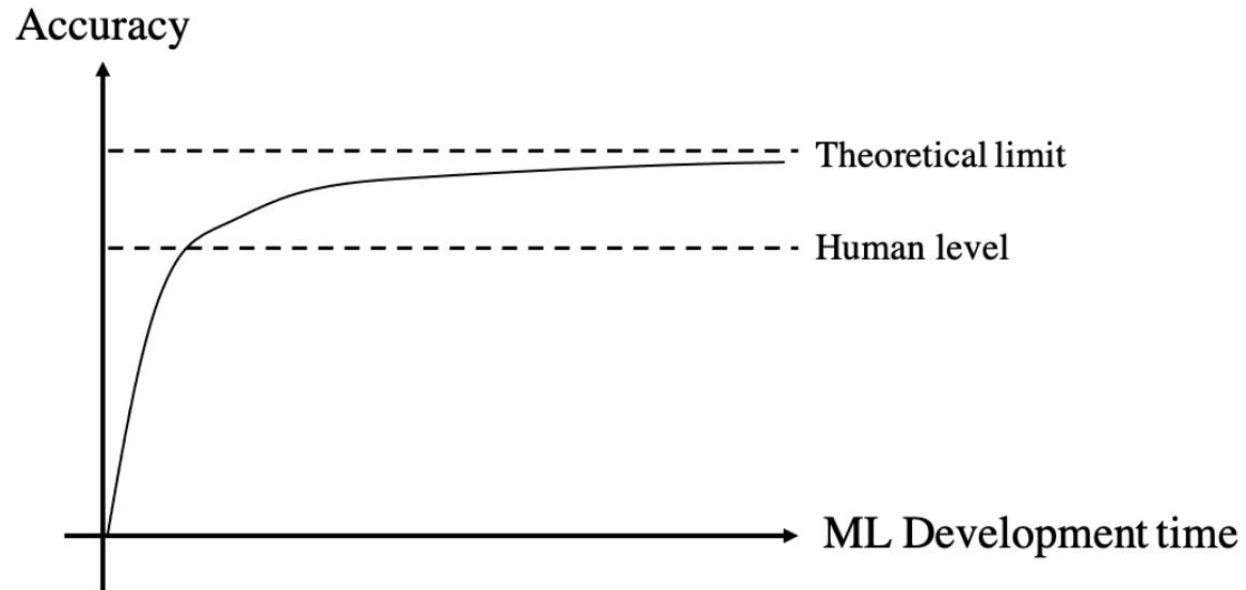


Capacity = size of the model

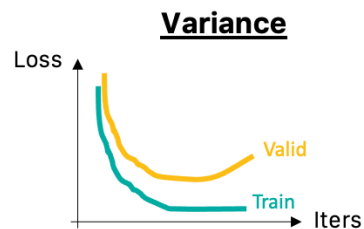
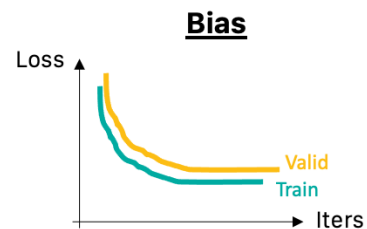
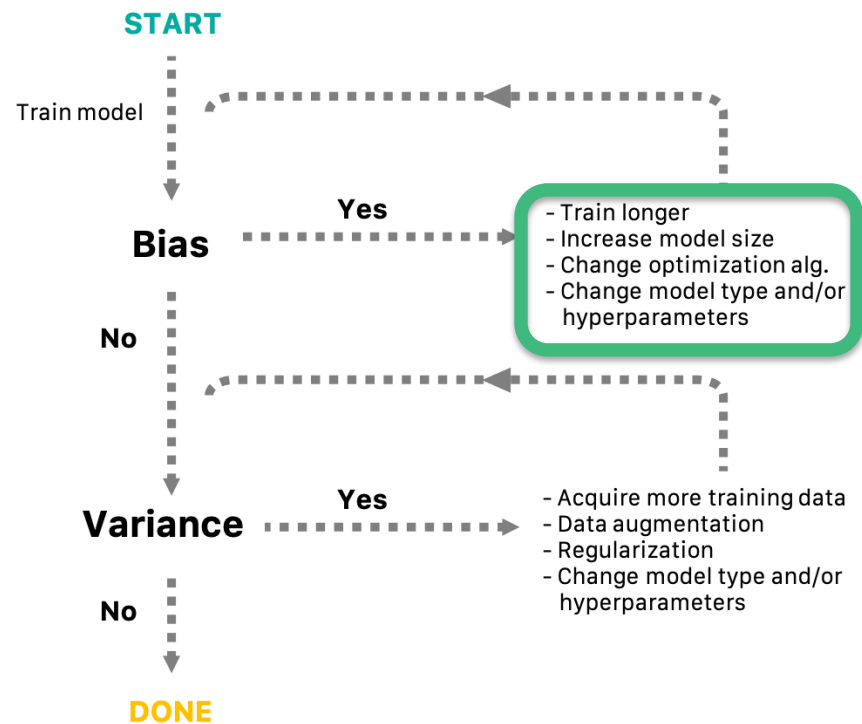
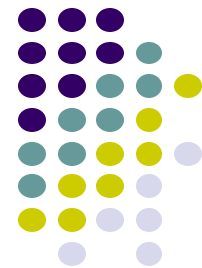
Expectations from ML model



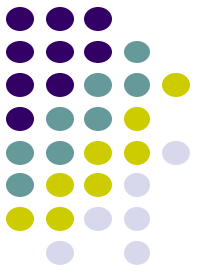
Always set expectations ahead of training!



Summary



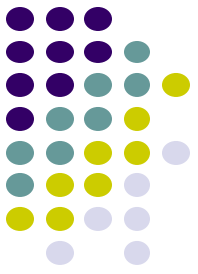
ML hyperparameters



Hyperparameter: *parameter of a learning algorithm that is **not learned**, but chosen upfront*

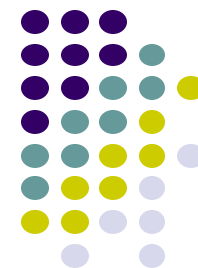
- model size (capacity)
- activation function;
- optimizer,
- learning rate
- additional optimizer hyperparameters,
- batch size...

Early Stopping



It involves stopping the training process before the model fully converges on the training data, preventing overfitting.

Regularization

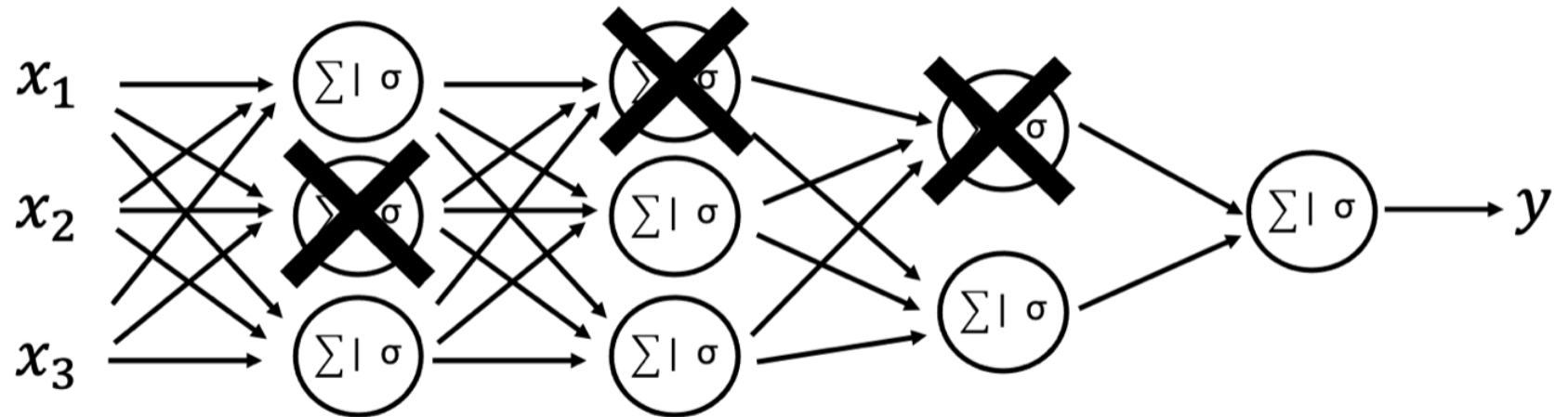
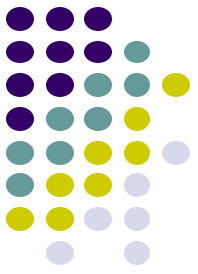


ML problem theory: *to avoid overfitting, add prior knowledge of the parameters to the loss function*

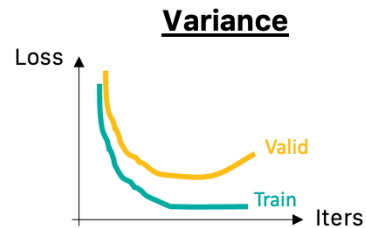
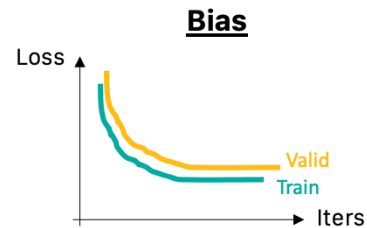
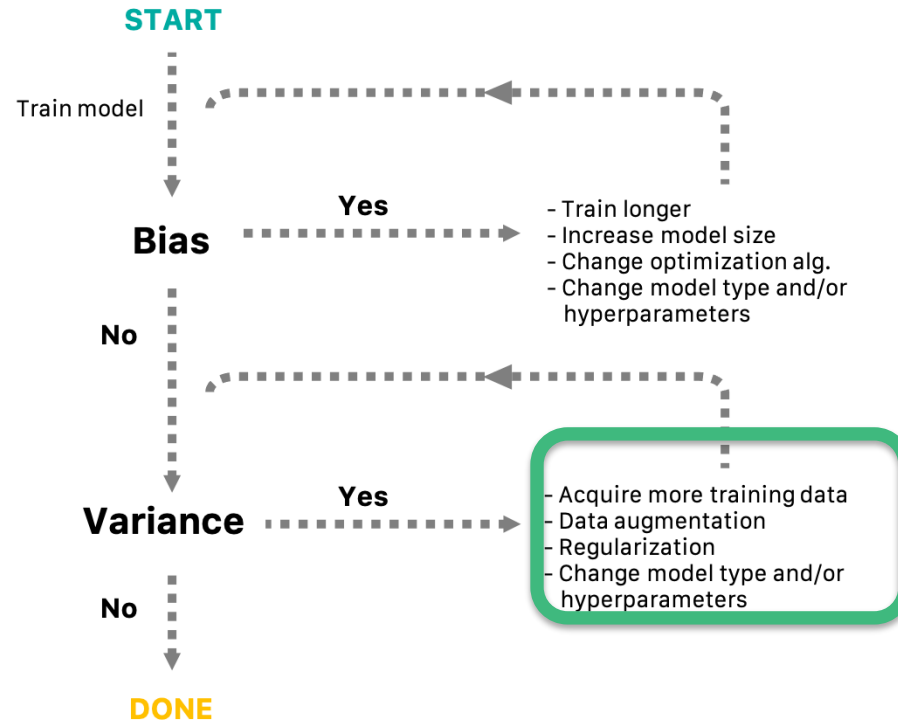
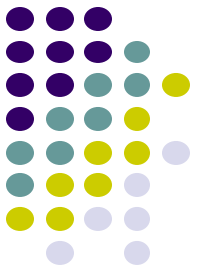
$$J_{\theta} = \frac{1}{N_s} \sum_{i=1}^{N_s} \|y^{(i)} - f_{\theta}(\mathbf{x}^{(i)})\|_2^2 + \lambda R(\theta)$$

- *L2 regularization or weight decay*: $\|\boldsymbol{\theta}\|_2^2 = \boldsymbol{\theta}^T \boldsymbol{\theta}$. Ensures small Euclidean norm for the free-parameters;
- *L1 regularization*: $\|\boldsymbol{\theta}\|_1$. Ensures small L1 norm for the free-parameters. By favouring sparsity in the parameters of the model, this regularization can allow compressing the trained model by storing only the non-negative parameters and their indices.

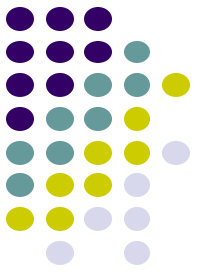
Regularization - dropout



Summary



Data augmentation



Availability of large set of trustworthy training data = key to success of ML

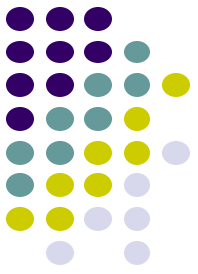
In the presence of few training data, apply transformations to enrich the dataset



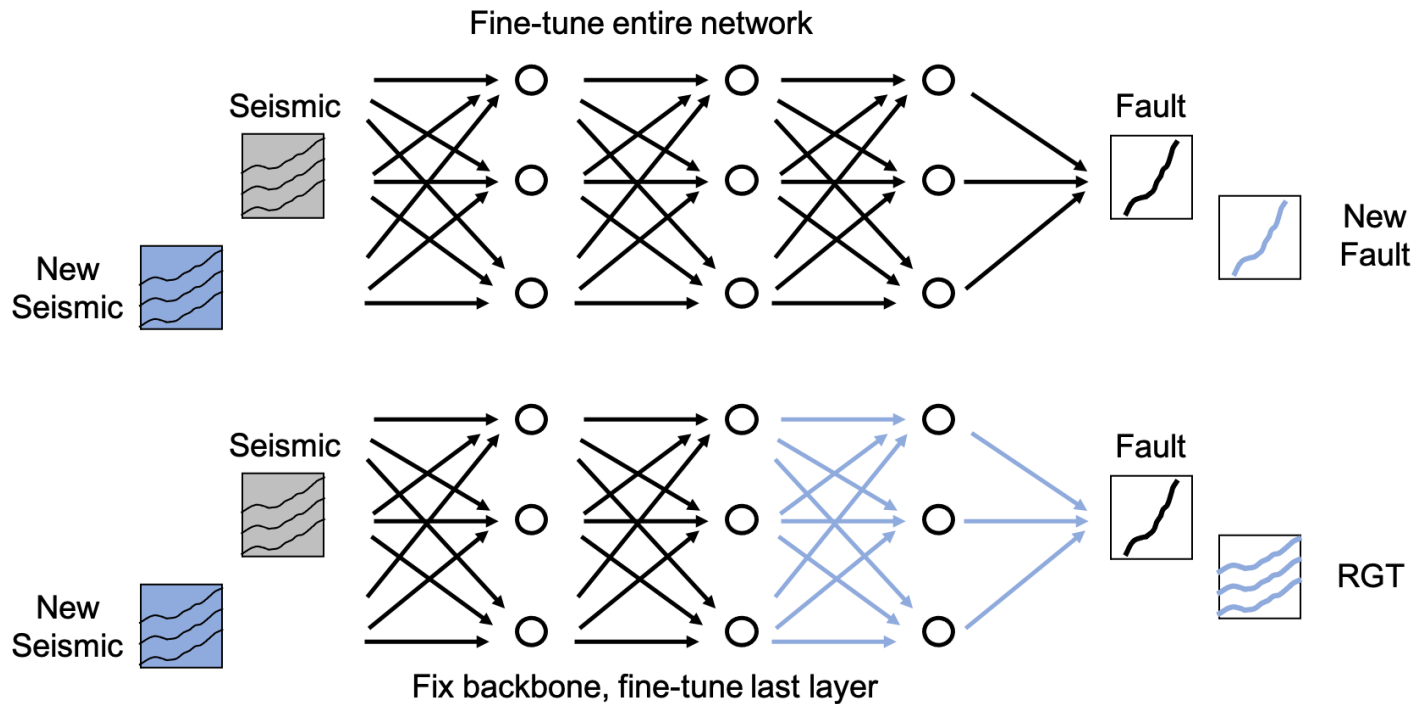
Ex: images

- cropping,
- flipping,
- rotating,
- shearing,
- averaging,
- color shifting

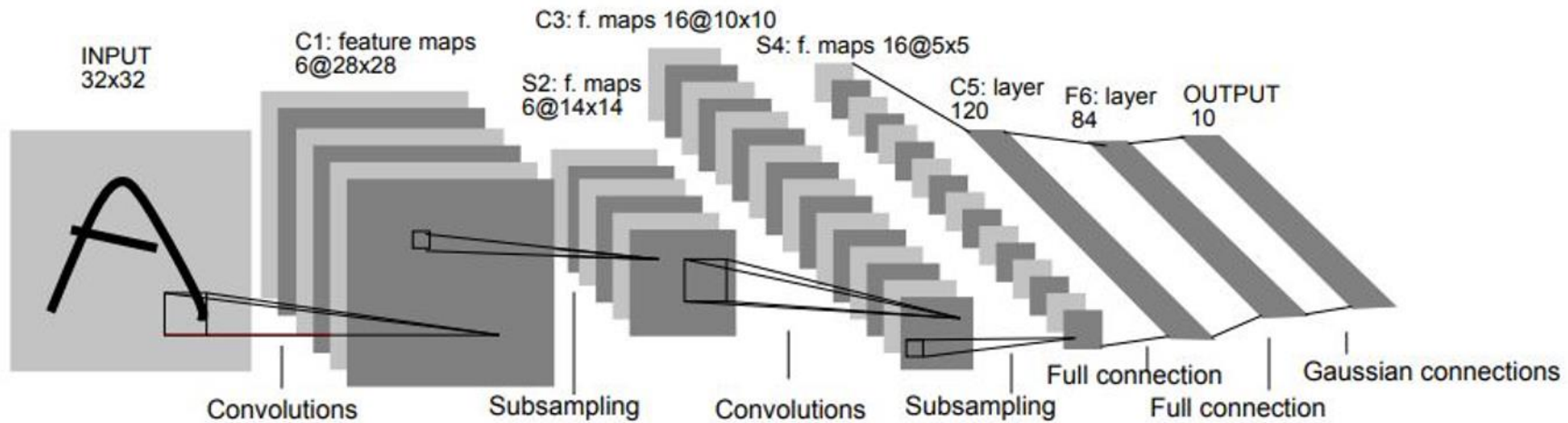
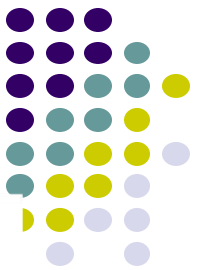
Transfer learning



Special case of data augmentation: use model trained in other data

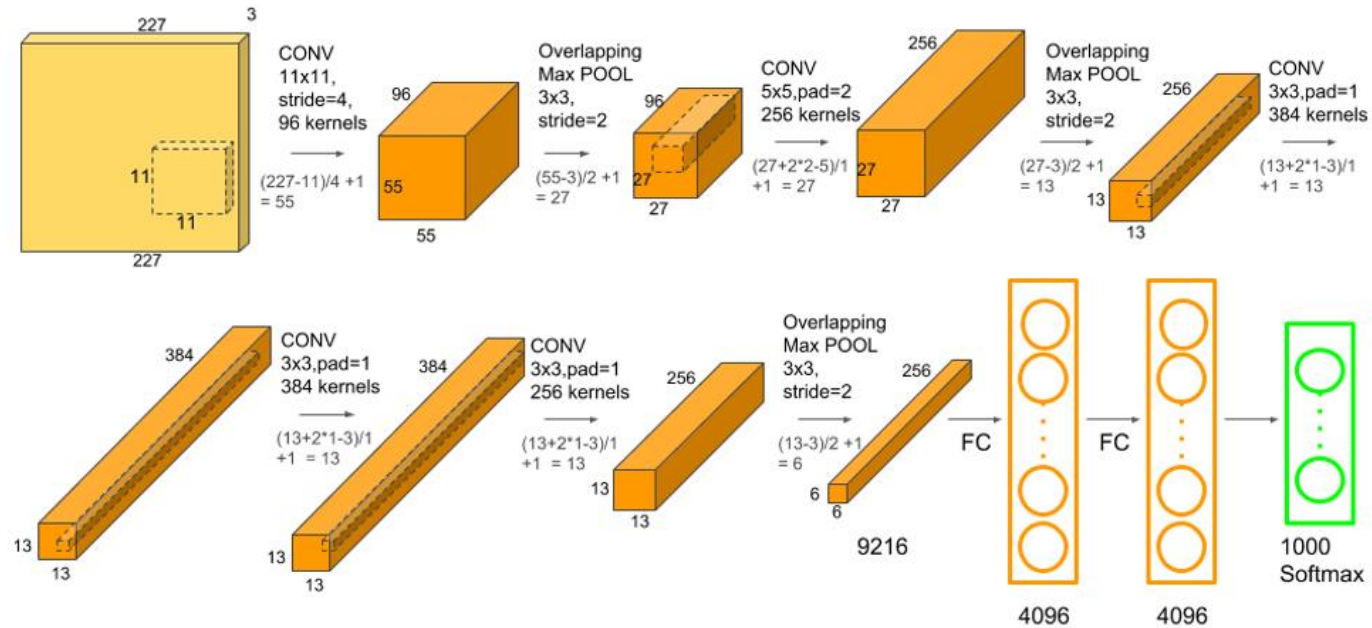
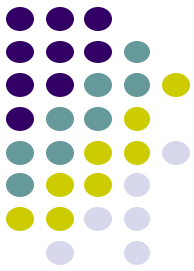


LeNet-5



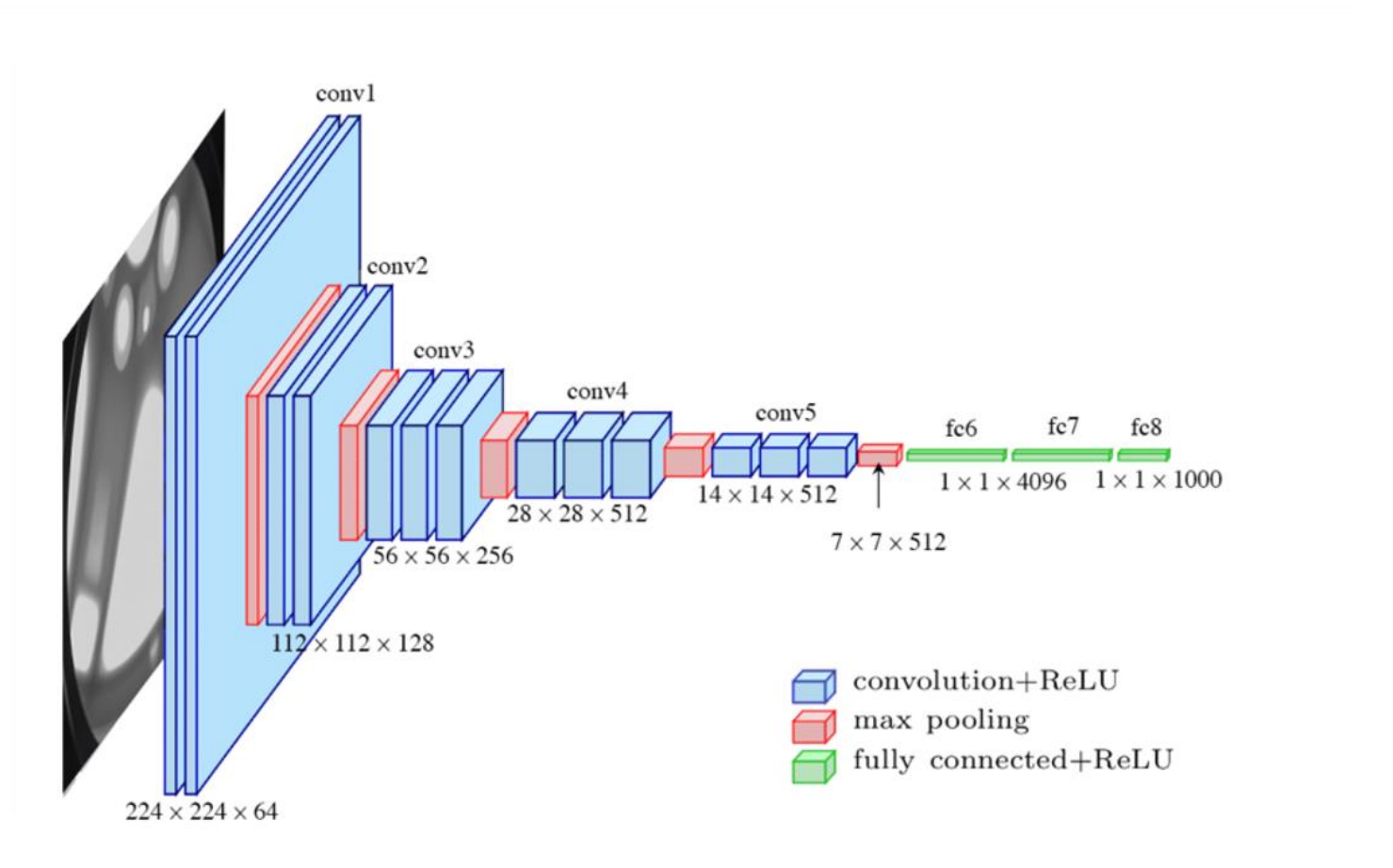
Source: <https://paperswithcode.com/method/lenet>

AlexNet

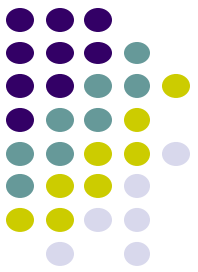


Source: <https://neurohive.io/en/popular-networks/alexnet-imagenet-classification-with-deep-convolutional-neural-networks/>

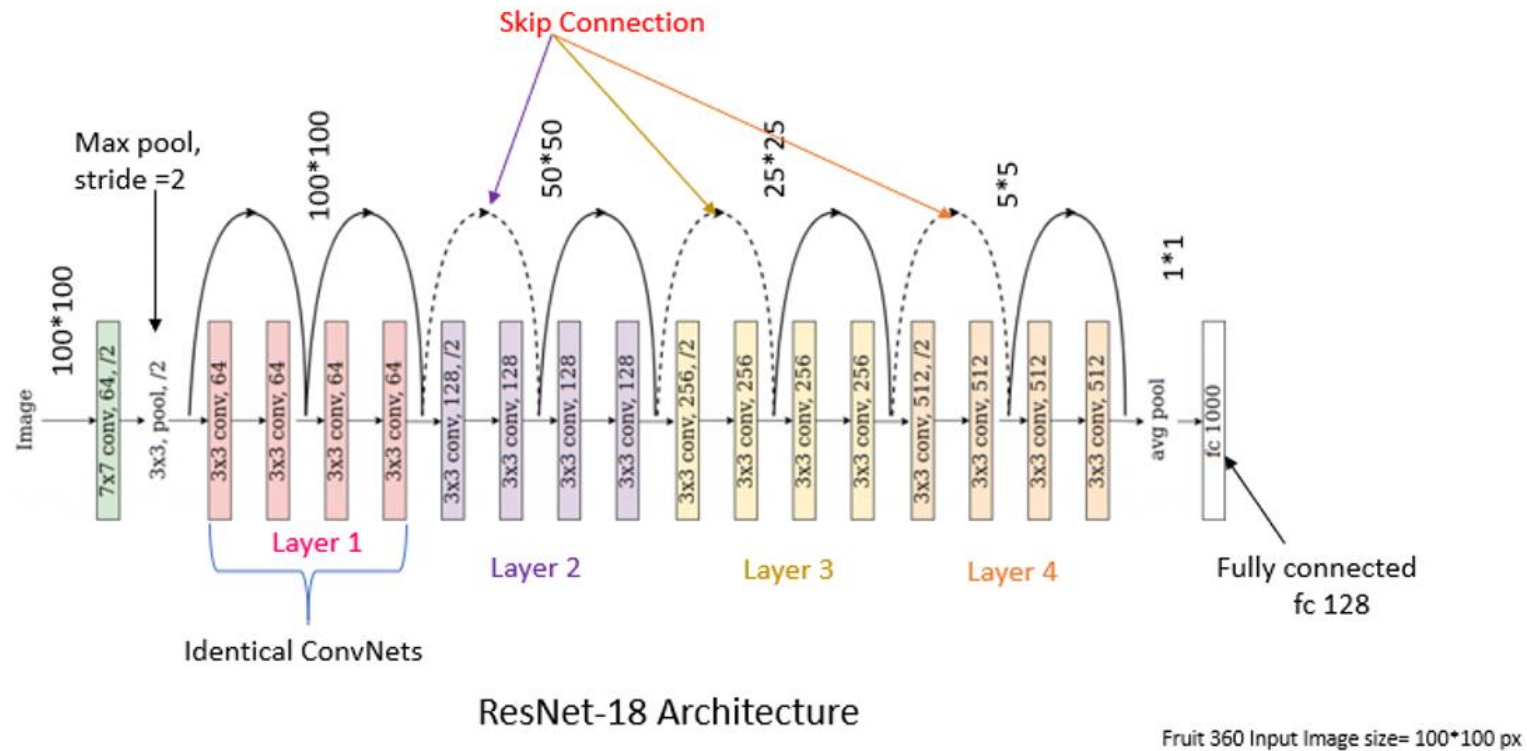
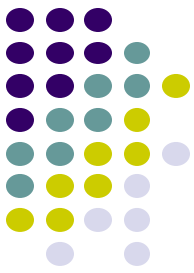
VGG



Source: https://www.researchgate.net/publication/322512435_Automatic_localization_of_casting_defects_with_convolutional_neural_networks

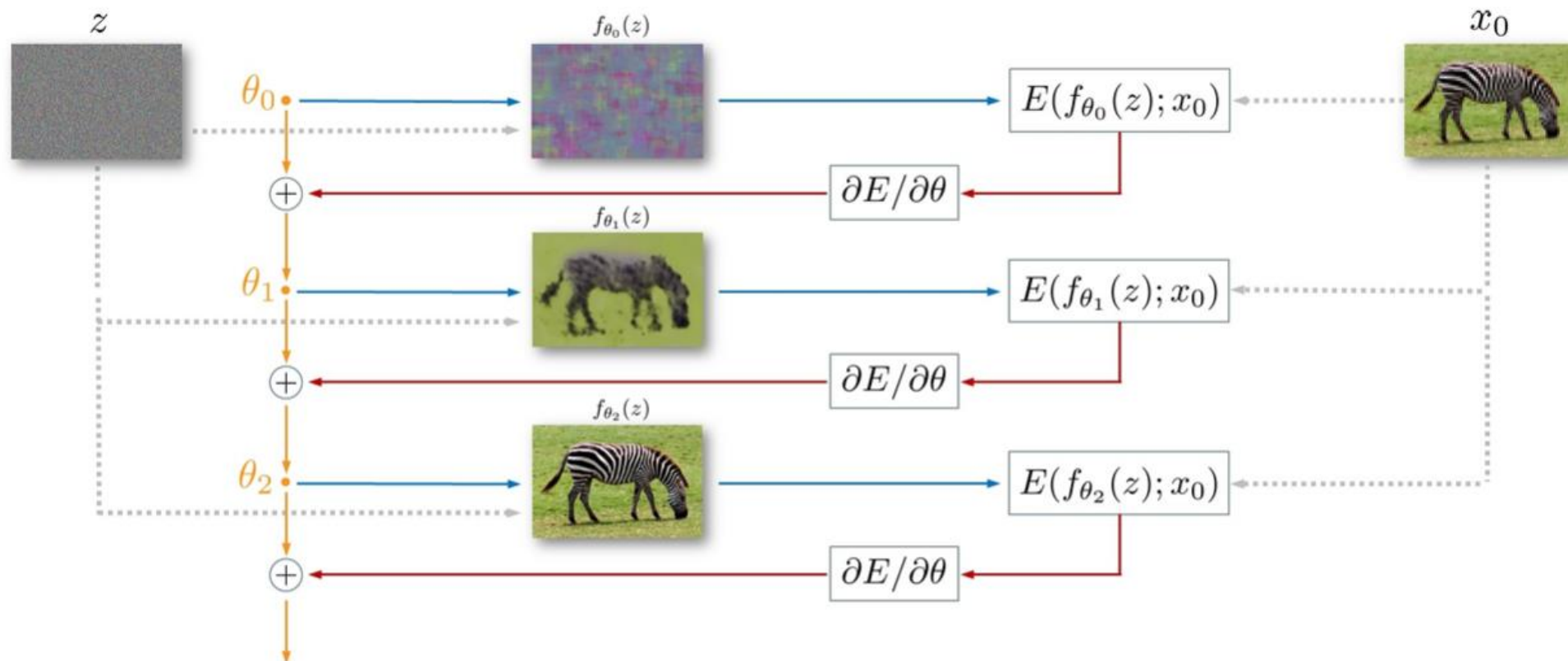
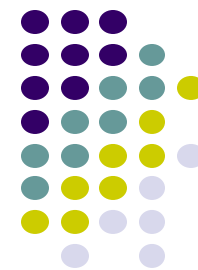


ResNet



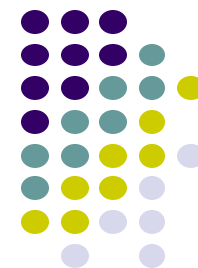
Source: <https://www.pluralsight.com/guides/introduction-to-resnet>

Deep Image Prior (DIP)



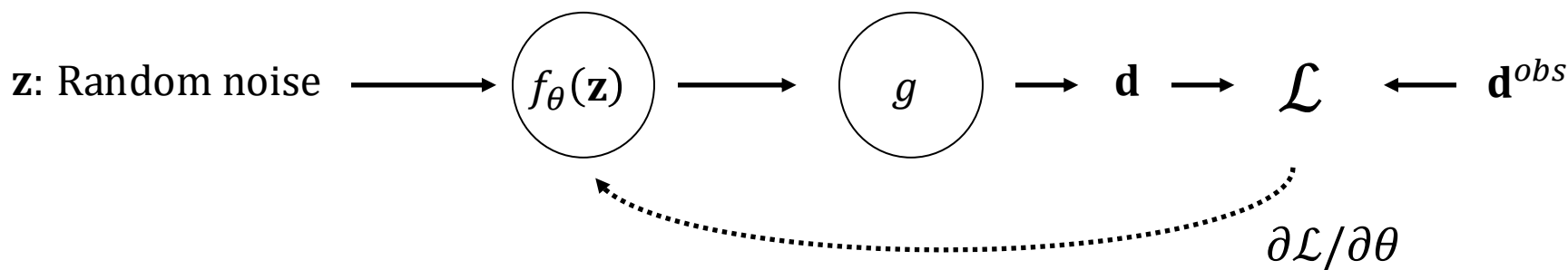
Ulyanov, D., Vedaldi, A., & Lempitsky, V. (2018). Deep image prior. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 9446-9454).

Deep Image Prior (DIP)

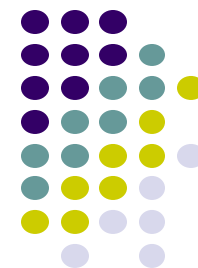


Given a suitable neural network architecture, a set of solutions can be promoted (and the rest rejected)

$$\text{DIP: } J = \|\mathbf{d}^{obs} - g(f_{\theta}(\mathbf{z}))\|_p^p$$



Deep Image Prior (DIP)



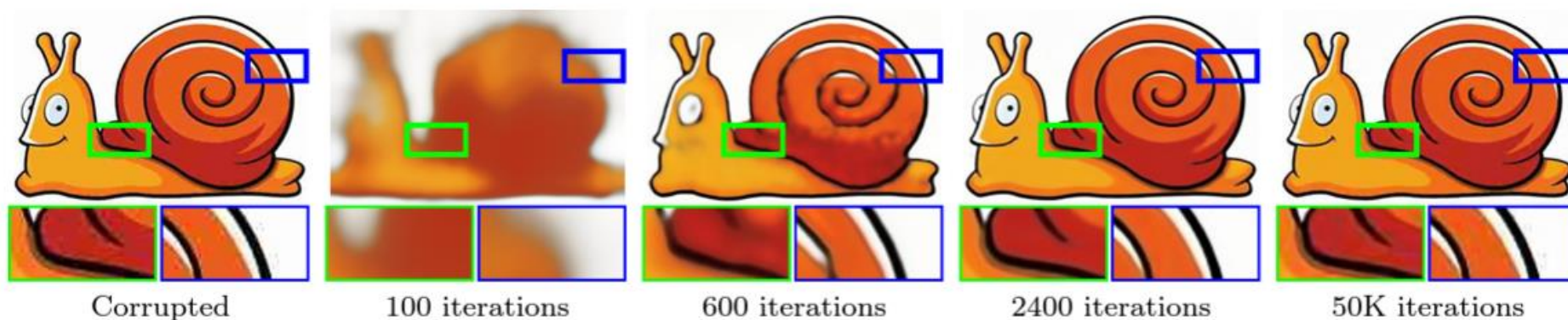
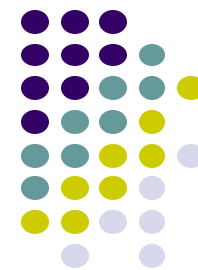
Benefits:

- Fully unsupervised, does not require any training sample
- Very easy to implement

Limitations:

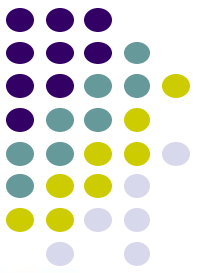
- Requires a user to define a suitable network architecture (like for traditional hand-crafted priors)
- Slow convergence

Deep Image Prior (DIP)

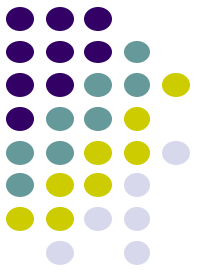


Ulyanov, D., Vedaldi, A., & Lempitsky, V. (2018). Deep image prior. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 9446-9454).

Deep Image Prior (DIP)

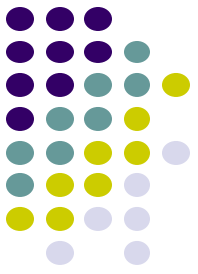


Ulyanov, D., Vedaldi, A., & Lempitsky, V. (2018). Deep image prior. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 9446-9454).

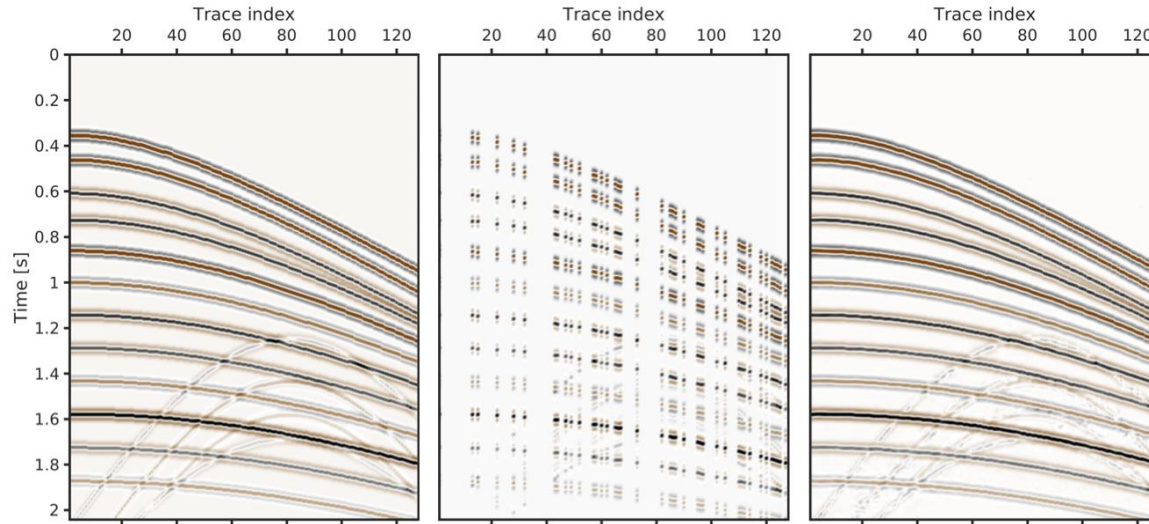


Application

Deep Image Prior (DIP)



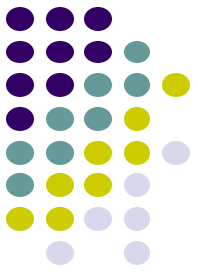
Examples in geophysics



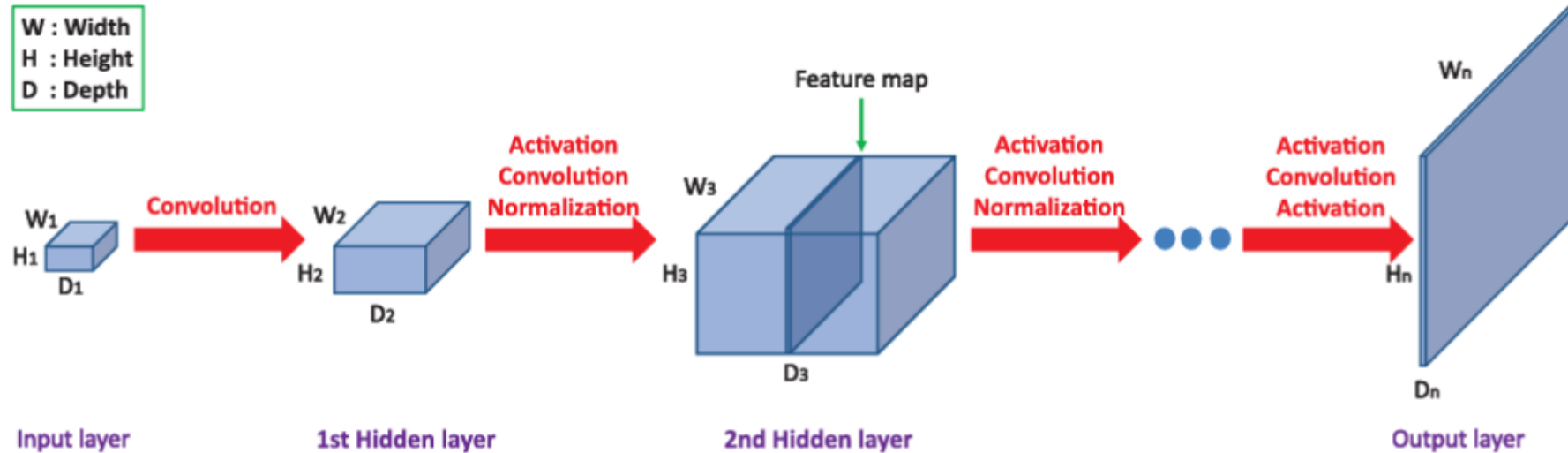
CNNs have natural interpolation capabilities and reject noise

Source: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9310238>

Deep Image Prior (DIP)

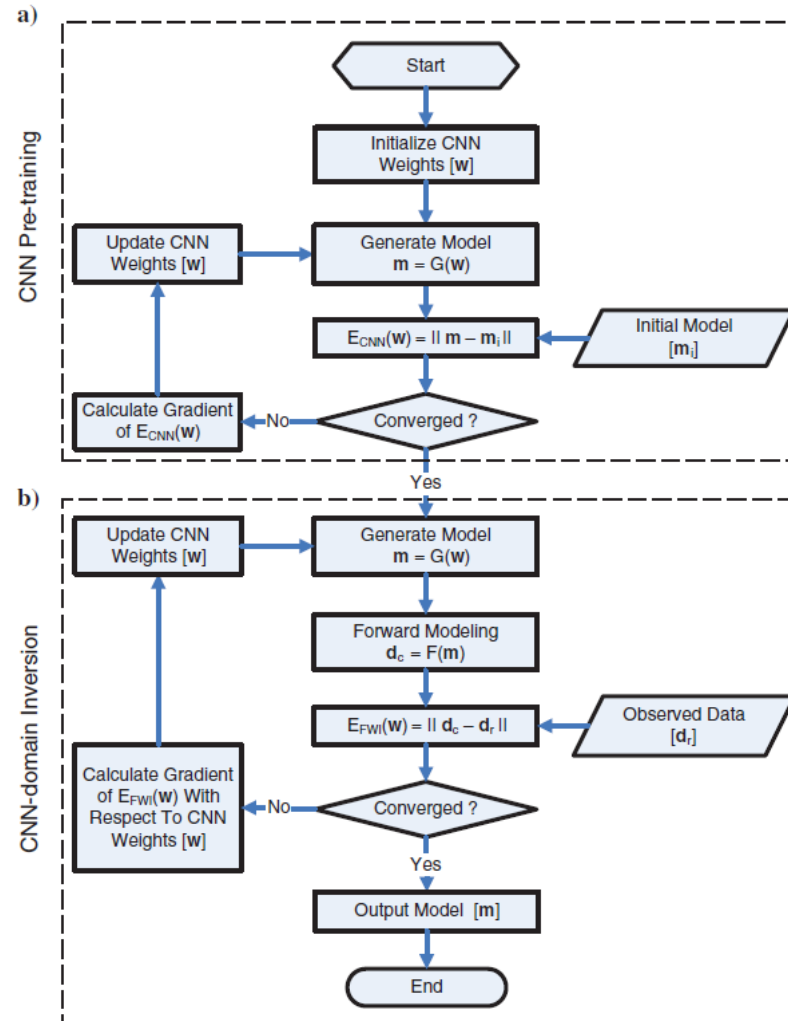


Examples in geophysics

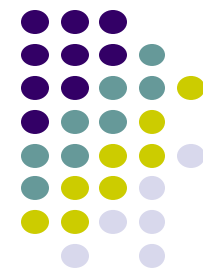


Wu, Y., & McMechan, G. A. (2019). Parametric convolutional neural network-domain full-waveform inversion. *Geophysics*, 84(6), R881-R896.

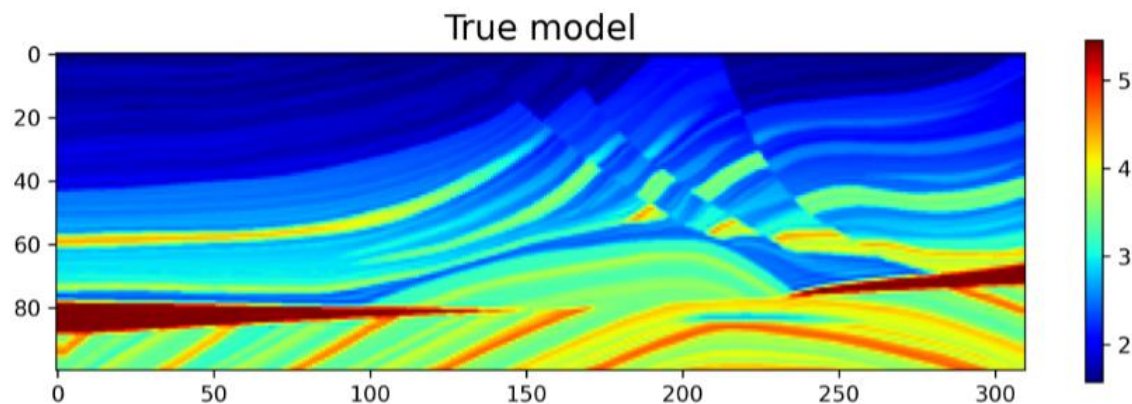
Deep Image Prior (DIP)



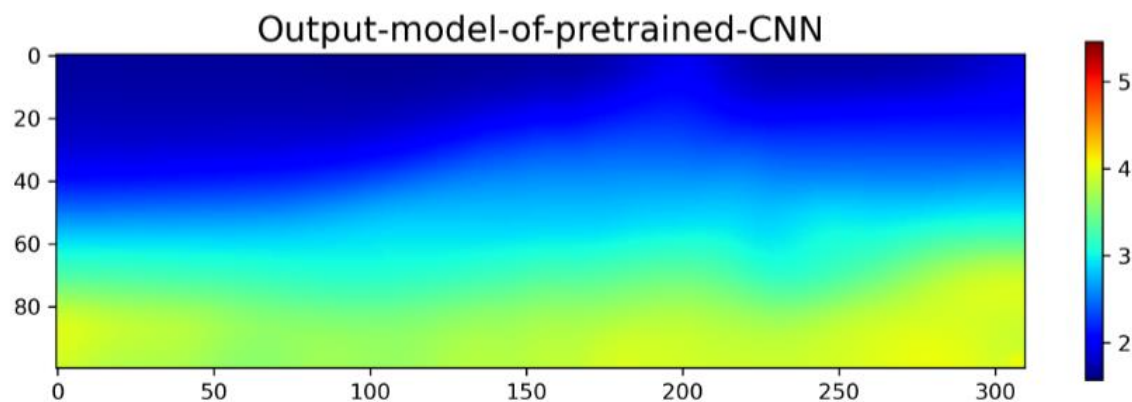
Deep Image Prior (DIP)



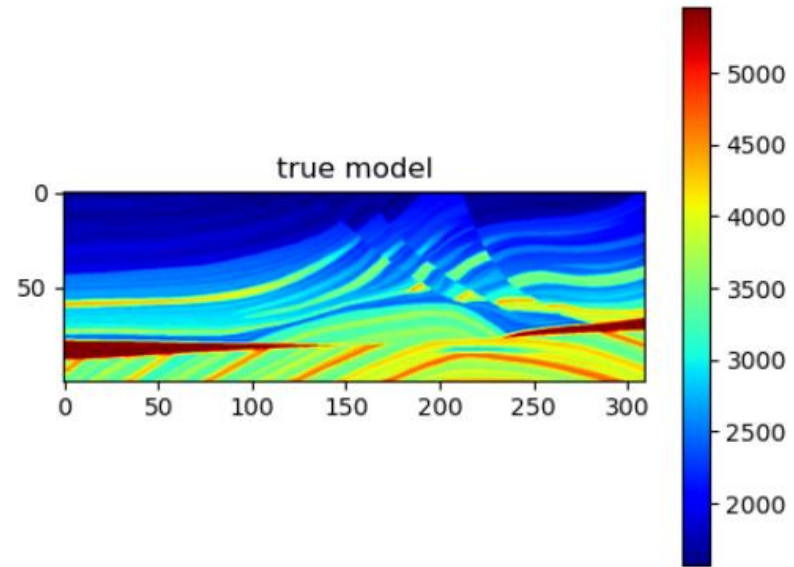
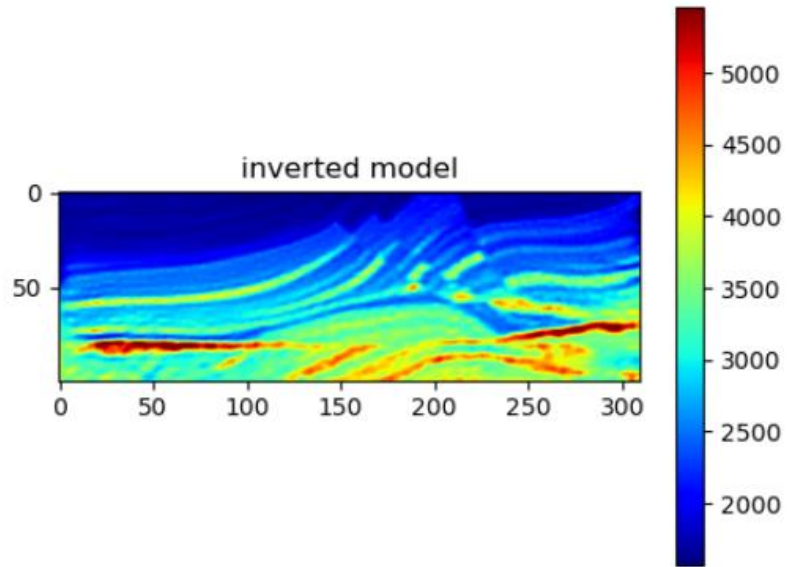
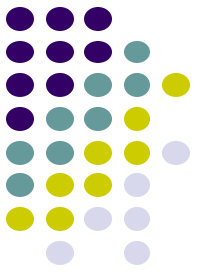
The true model



The initial model



Deep Image Prior (DIP)





Thank You

30 17:50